



AI PROJECT

PAXIS AI

Personalized AI Exploration and Intelligent Strategy

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**An AI-Powered System for Personalized Learning Journey Generation,
Resource Discovery, Progress Tracking and Next-Action Recommendation**

HCL Amplified Hackathon

AI-Powered Personalized Learning Solution

Department of Computer Science and Engineering

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Project Abstract

PAXIS AI, which stands for Personalized AI Exploration and Intelligent Strategy, is designed to help learners transform broad learning goals into personalized and structured learning journeys.

The system addresses a common problem in modern learning environments: learners have access to a large number of courses, videos, websites and learning materials, but often lack a clear understanding of where to begin, what to learn next, which resources to select and how to measure their progress.

PAXIS approaches this problem by considering the learner as well as the goal. The learner can provide information such as interests, current knowledge, previously learned topics, desired outcomes and available learning time. The learner can then describe the intended goal using natural language in English, Hindi or Hinglish.

The system uses this information to create a personalized learning journey consisting of structured stages. Each stage communicates what the learner should learn, practice and complete before moving forward. Relevant educational resources can also be associated with the stages.

PAXIS further extends the learning experience through roadmap tracking, skill development tracking, milestones, learning streaks, saved conversations and saved learning journeys. Progress information is used to help the learner identify an appropriate next action instead of repeatedly searching for what to study.

The overall objective is to transform learning from an unstructured collection of resources into a continuous process in which the learner has a defined starting point, a target outcome, a structured path, measurable progress and ongoing guidance.

1. Introduction

1.1 Background

The availability of online educational content has significantly increased the number of opportunities available to learners. A learner can find courses, tutorials, educational videos, documentation, articles and practical learning materials for almost every technical and professional domain.

However, the availability of information also creates a new challenge. Learners are often required to make several decisions independently:

- Where should learning begin?
- Which topics are essential?
- What should be learned first?
- Which topics depend on previously learned concepts?
- Which learning resources should be selected?
- How much progress has been made?
- What should be done after completing the current topic?

A conventional course recommendation system can help identify relevant content, but relevance alone does not necessarily provide a complete learning strategy.

PAXIS AI focuses on the larger problem of constructing and continuing a personalized learning journey.

1.2 Overview of PAXIS AI

PAXIS AI is designed around a simple principle:

The learner provides the destination; PAXIS helps construct the path.

The learner begins by providing information about their current learning state and then describes their intended goal.

PAXIS combines these two pieces of information to produce a personalized journey.

The system considers:

- Learner interests.
- Current knowledge.
- Previously learned concepts.
- Desired outcome.
- Available learning time.

- Current progress.

This means that the same goal can result in different journeys for different learners.

1.3 Problem Understanding

The central problem is not the lack of educational information. Instead, the problem is the difficulty of converting available information into a useful sequence of actions.

A learner may know the final goal but still be uncertain about the intermediate steps.

For example, a learner who wants to become a Python Developer may need to determine:

- Which programming fundamentals are required.
- Which Python concepts should be studied.
- Which concepts should be practiced.
- Which development skills should follow.
- Which resources are suitable.
- When the learner is ready to move to the next stage.

PAXIS addresses this decision-making problem through a personalized, stage-based learning journey.

1.4 Problem Statement

The project addresses the following major issues:

- Generic learning paths do not account for individual differences between learners.
- Learners with different levels of prior knowledge may receive the same recommendations.
- Learners may spend significant time searching for appropriate educational resources.
- Broad learning goals are difficult to convert into actionable steps.
- Learners may lose track of their progress across multiple learning activities.
- After completing one topic, learners may not know which action should follow.
- Learning activity and previous conversations may be lost when a learner returns later.

1.5 Objectives

The objectives of PAXIS AI are:

1. To understand the learner's current learning context.
2. To accept learning goals through natural-language interaction.

3. To support English, Hindi and Hinglish communication.
4. To create personalized learning journeys.
5. To divide large goals into manageable stages.
6. To provide learning and practice guidance for each stage.
7. To identify useful learning resources.
8. To maintain roadmap and stage completion information.
9. To track skill development and milestones.
10. To maintain learning streak information.
11. To recommend the next useful learning action.
12. To preserve conversations and journeys for future sessions.

1.6 Scope

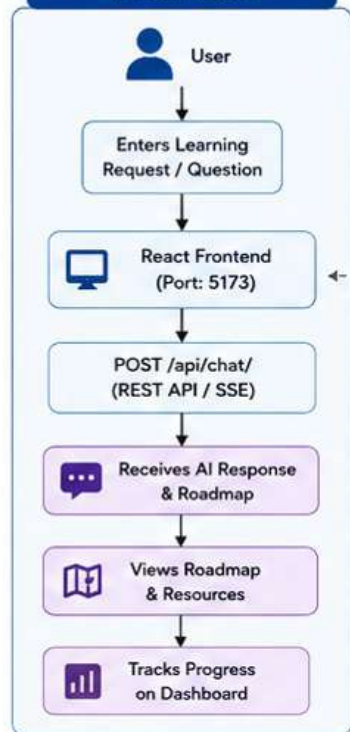
The scope of the project includes the complete learner journey from goal creation to continued progress.

The major areas within scope are:

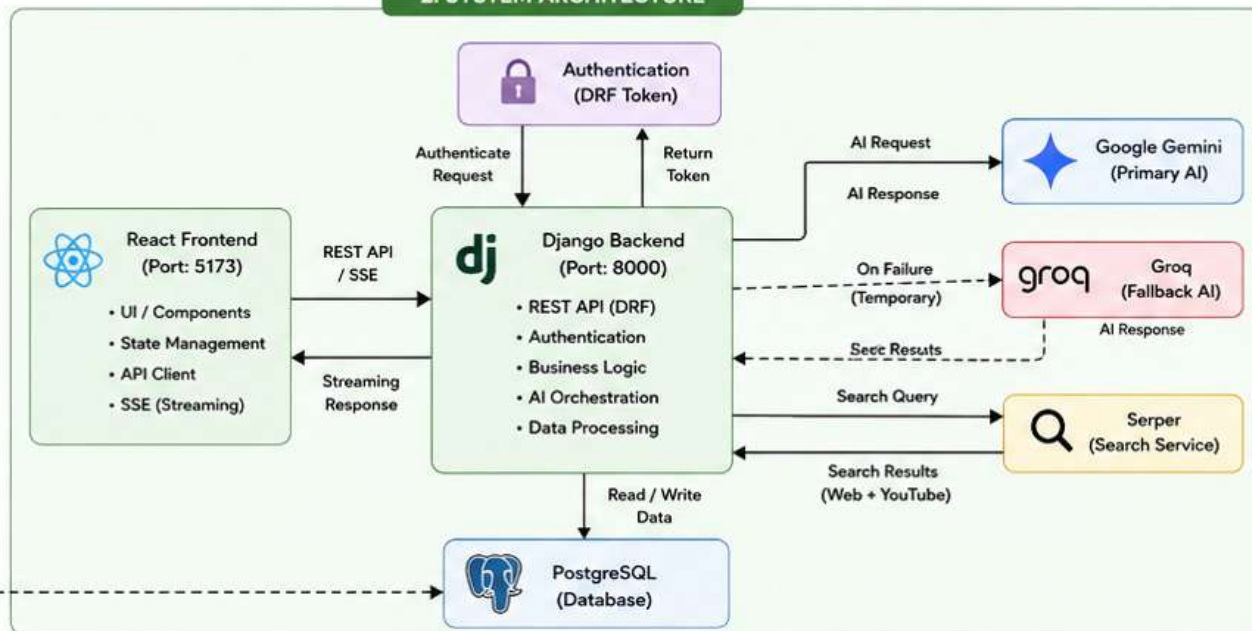
- Learner profiling.
- Natural-language goal input.
- Goal and learner understanding.
- Personalized roadmap generation.
- Learning resource discovery.
- Stage completion.
- Progress tracking.
- Skill tracking.
- Milestone management.
- Learning streaks.
- Next-action recommendation.
- Saved conversations.
- Saved learning journeys.
- Downloadable learning journeys.

PAXIS AI – PROJECT FLOWCHART

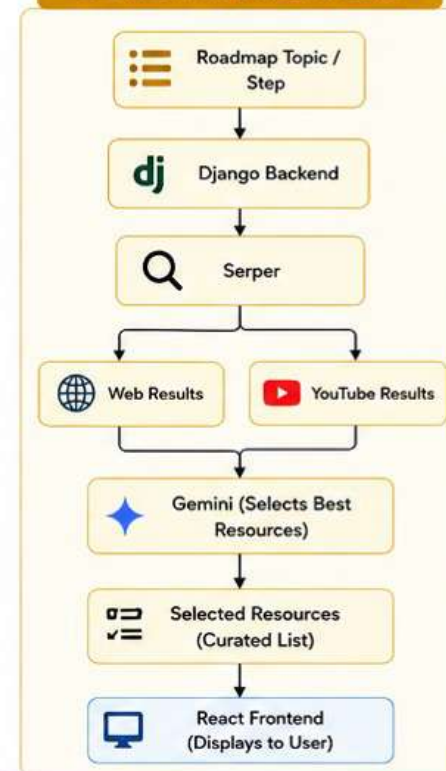
1. USER FLOW



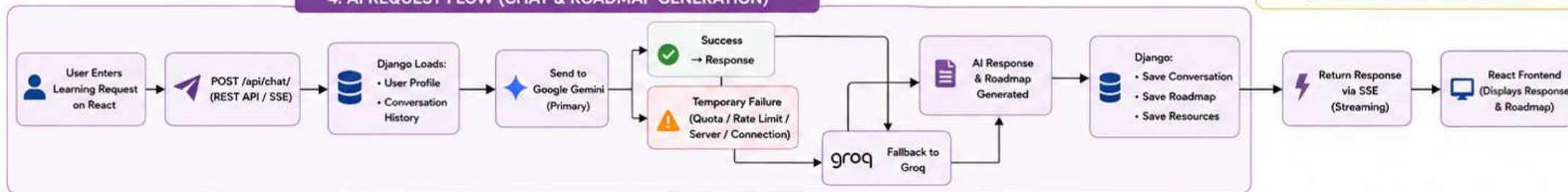
2. SYSTEM ARCHITECTURE



3. STUDY MATERIAL PIPELINE



4. AI REQUEST FLOW (CHAT & ROADMAP GENERATION)



5. DATA STORAGE (POSTGRESQL)



6. KEY FEATURES FLOW



TECHNOLOGY STACK



Frontend: React, Vite, React Router, JavaScript, CSS



Backend: Python, Django, Django REST Framework



AI: Google Gemini, Groq (Fallback)



Search: Serper (Web & YouTube)



Database: PostgreSQL (Django ORM)



Auth: DRF Token Authentication

2. Existing Approach and Proposed Solution

2.1 Limitations of Conventional Learning

Traditional online learning generally provides a large collection of resources. Although these resources may be useful individually, the learner is often responsible for determining the correct sequence.

This creates several difficulties:

- Resource overload.
- Lack of individual personalization.
- Difficulty identifying prerequisites.
- Lack of continuous progress guidance.
- Repeated searching for the next topic.
- Limited connection between learning goals and daily actions.

2.2 Need for a Personalized Approach

A learning journey should reflect the learner's current state.

Consider two learners with the same goal of becoming Python Developers.

Learner	Learning Context
Learner A	Complete beginner with limited programming knowledge.
Learner B	Already understands programming fundamentals.

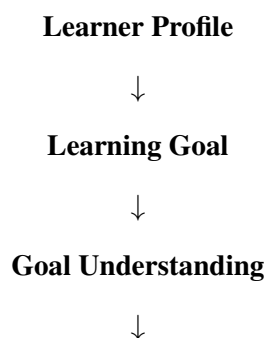
Table 2.1: Different Starting Conditions

Giving both learners exactly the same path would ignore important differences in their starting points.

PAXIS therefore combines learner context with the target goal before constructing the journey.

2.3 Proposed Solution

The proposed solution consists of a continuous personalized learning cycle.



Learner–Goal Personalization



Learning Journey Generation



Resource Discovery



Learn and Practice



Progress Update



Next Action



Continue Journey

2.4 Advantages of the Proposed Approach

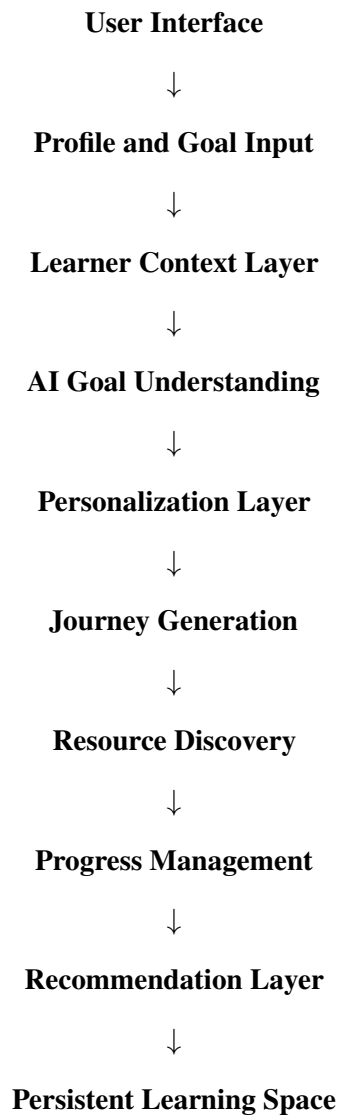
The proposed approach provides:

- A learner-specific starting point.
- A structured sequence instead of an unorganized list.
- Goal-oriented learning.
- Context-aware recommendations.
- Progress visibility.
- Continuous next-step guidance.
- Persistent learning journeys.

3. System Architecture

3.1 Architecture Overview

PAXIS AI can be represented as a set of interconnected logical components.



3.2 User Interface Layer

The user interface acts as the interaction point between the learner and PAXIS.

The learner can:

- Create or provide their profile.
- Enter learning goals.
- View generated journeys.
- Open learning resources.

- Track completed stages.
- View progress.
- Continue saved journeys.

3.3 Learner Context Layer

This layer represents the information that describes the learner.

The context includes:

- Interests.
- Current knowledge.
- Previous learning.
- Desired outcome.
- Available weekly learning time.

The learner context becomes an important input to personalization.

3.4 Goal Understanding Layer

The learner can describe the desired outcome using natural language.

For example:

“Mujhe 6 mahine mein Python Developer banna hai.”

The system uses the stated goal together with learner context instead of treating the goal as an isolated text input.

3.5 Personalization Layer

The personalization layer connects the learner’s starting state with the intended destination.

Its logical inputs include:

- Current knowledge.
- Previous learning.
- Interests.
- Learning goal.
- Available learning time.

The output is a learner-specific learning strategy.

3.6 Learning Journey Generator

The journey generator converts the broad objective into stages.

Each stage identifies:

- What should be learned.
- What should be practiced.
- What should be completed.
- What follows the current stage.
- Which milestones are relevant.

3.7 Resource Layer

The resource layer provides useful educational material for learning stages.

Potential resources include:

- Websites.
- Educational videos.
- Learning materials.

The journey remains available even when an appropriate resource cannot be found.

3.8 Progress Layer

The progress layer maintains the learner's current state.

It represents:

- Stage completion.
- Overall completion.
- Skill progress.
- Milestones.
- Recent activity.
- Learning streak.

3.9 Recommendation Layer

The recommendation layer uses the learner's current progress to provide guidance about the next useful action.

This creates continuity between stages instead of requiring the learner to restart the search process after every completed activity.

4. AI and Machine Learning Approach

4.1 Role of AI in PAXIS

Artificial intelligence forms the reasoning layer of PAXIS AI.

The system is designed to understand learner requirements expressed in natural language and transform those requirements into a structured learning strategy.

The AI-related responsibilities include:

- Understanding learning goals.
- Interpreting learner context.
- Connecting the learner's current state with the target outcome.
- Generating an appropriate sequence of learning stages.
- Supporting conversational interaction.
- Providing context-aware next-action guidance.

4.2 Natural Language Processing

Natural Language Processing (NLP) is relevant to PAXIS because users are not required to communicate through a fixed command structure.

A learner can describe an objective in ordinary language.

For example:

"I want to become a Python Developer in six months."

or:

"Mujhe 6 mahine mein Python Developer banna hai."

The natural-language layer therefore needs to identify the intended learning objective and preserve its meaning for the subsequent personalization process.

4.3 Learner Profiling

Learner profiling is an important component of personalization.

The learner profile provides contextual attributes such as:

Profile Attribute	Purpose
Interests	Provides information about learner preferences and context.
Knowledge Level	Helps identify an appropriate starting point.
Previous Learning	Helps identify concepts already covered by the learner.
Goal	Defines the desired destination.
Available Time	Provides a practical learning constraint.

Table 4.1: Learner Profile Attributes

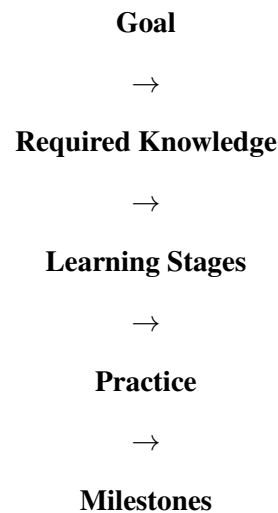
4.4 Goal-to-Skill Reasoning

A broad goal generally requires multiple intermediate skills.

For example, the goal of becoming a Python Developer can involve programming fundamentals, Python concepts, development practices and other supporting skills.

PAXIS approaches the goal as a destination and divides the required learning into smaller stages.

The general reasoning process is:



4.5 Context-Aware Personalization

The key distinction between ordinary recommendation and PAXIS is the use of learner context.

The conceptual personalization function can be represented as:

$$J = f(G, K, I, P, T)$$

where:

- J = Personalized learning journey.
- G = Learning goal.
- K = Current knowledge.

- I = Learner interests.
- P = Previously learned concepts.
- T = Available learning time.

The expression represents the design principle of PAXIS rather than a claim about a specific mathematical model implemented in the system.

4.6 Progress-Aware Recommendation

After the journey has been generated, learner progress becomes another important input.

The recommendation process can be represented as:

$$A_{next} = f(S_{current}, C, M, G)$$

where:

- A_{next} = Recommended next action.
- $S_{current}$ = Current learning stage.
- C = Completed activities.
- M = Remaining milestones.
- G = Original learning goal.

This design allows the recommendation to remain connected to the learner's actual position.

4.7 Multilingual Interaction

PAXIS is designed to support:

- English.
- Hindi.
- Hinglish.

This enables learners to express their goals in a more natural manner without requiring a strict language or command format.

4.8 Important Implementation Note

The available project material establishes the use of AI-oriented goal understanding, personalization, recommendation and conversational interaction. It does not specify a particular machine-learning model, training dataset, embedding model or algorithm.

Therefore, this report does not attribute an unsupported model such as BERT, TF-IDF, cosine similarity, RAG or a particular large language model to the implementation.

If a specific model is implemented in the actual source code, it can be documented in this section together with its inputs, outputs, parameters and evaluation results.

5. Core Modules and Features

5.1 Profile Creation

The learner begins by creating a personal profile.

The profile captures information required for personalization.

- Interests.
- Current knowledge level.
- Previously learned topics.
- Intended outcomes.
- Available learning time.

5.2 Natural-Language Goal Input

The learner can describe what they want to achieve without following special commands.

This reduces the interaction barrier and allows the learner to focus on the intended outcome.

5.3 Personalized Journey

PAXIS produces a structured learning journey based on learner context and goal.

The journey includes:

- Learning stages.
- Practice requirements.
- Completion criteria.
- Upcoming stages.
- Milestones.

5.4 Learning Resources

Resources can be associated with the different stages of the journey.

The system can provide:

- Helpful websites.
- Educational videos.
- Learning materials.

5.5 Roadmap Tracking

A learner can choose:

Track This Roadmap

The selected journey is then stored in the learner's personal learning space.

5.6 Stage Completion

Learners can mark stages according to their current state.

Stage	Status
Stage 1	Completed ✓
Stage 2	Completed ✓
Stage 3	In Progress
Stage 4	Not Started

Table 5.1: Example of Stage Tracking

5.7 Progress Dashboard

The learner can monitor:

- Overall completion.
- Skill development.
- Completed milestones.
- Recent activity.
- Learning streak.
- Current position.
- Recommended next action.

5.8 Milestones

Milestones divide a long-term objective into meaningful achievement points.

They provide learners with visible indicators of progress toward the final goal.

5.9 Learning Streaks

Learning streaks represent continued learning activity and provide learners with an additional progress indicator.

5.10 Saved Conversations

Previous conversations are retained so that the learner can return to the learning process later.

5.11 Saved Learning Journeys

A generated journey can be saved so that the learner does not have to generate it again when returning to the system.

5.12 Downloadable Journey

The learning journey can be downloaded, allowing the learner to retain the roadmap for later reference.

6. Complete System Workflow

6.1 End-to-End Workflow

The complete PAXIS workflow consists of the following stages.

1. **Create Profile**

The learner provides information describing their current learning situation.

2. **Enter Goal**

The learner describes the intended outcome using natural language.

3. **Understand Learner**

PAXIS considers the learner's knowledge, interests, previous learning and available time.

4. **Understand Goal**

The system determines the intended learning destination.

5. **Generate Journey**

The goal is converted into a structured sequence of learning stages.

6. **Find Resources**

Useful learning resources can be associated with appropriate stages.

7. **Track Roadmap**

The learner can save the journey and begin following it.

8. **Complete Stages**

Learners mark activities and stages as they progress.

9. **Update Progress**

Completion, skills, milestones, activity and streak information are updated.

10. **Recommend Next Action**

The system identifies the next useful learning activity.

11. **Continue Journey**

The learner can return later and continue from the saved state.

6.2 Workflow Representation

YOUR GOAL



Tell PAXIS What You Want



PAXIS Understands Your Needs



Personalized Learning Journey



Learn + Practice



Complete Stages



Track Progress



Reach Milestones



Get Next Action



Achieve Goal

6.3 Journey Continuation

The workflow does not terminate when the initial roadmap is generated.

Instead, the learner continues through a cycle:

Learn → Complete → Track → Recommend → Learn Again

This makes PAXIS a continuing learning system rather than a one-time roadmap generator.

7. Illustrative Use Case

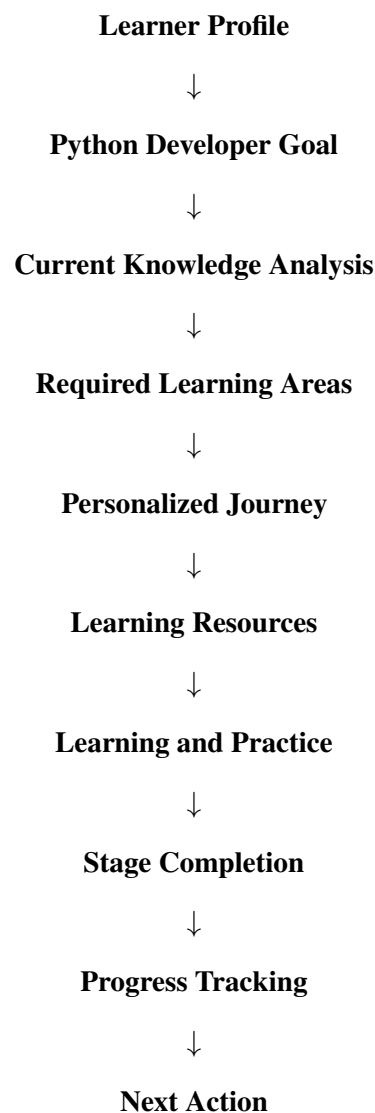
7.1 Python Developer Example

Consider a learner who wants to become a Python Developer within six months.

The learner enters:

“Mujhe 6 mahine mein Python Developer banna hai.”

PAXIS processes the request through the following logical sequence:



7.2 Personalization Example

Suppose Learner A is a complete beginner.

The journey can begin with foundational programming concepts.

Learner B may already understand programming fundamentals.

The journey can therefore begin from a later stage.

The difference demonstrates the central personalization principle:

Same Goal $\neq$ Same Learning Journey

7.3 Time-Based Personalization

Available learning time is another contextual factor.

For example:

Available Time	Possible Journey Characteristic
2 hours/week	More gradual progression
5 hours/week	Moderate progression
10 hours/week	Greater learning activity capacity

Table 7.1: Effect of Available Learning Time

The table illustrates the intended personalization principle rather than a fixed scheduling algorithm.

8. Challenges Faced

8.1 Understanding Broad Learning Goals

One of the major challenges is converting a broad statement into a structured learning objective.

A goal such as “become a Python Developer” does not directly specify all the knowledge and activities required.

The system therefore needs to transform the broad objective into smaller and more meaningful stages.

8.2 Different Starting Levels

Learners begin from different levels of knowledge.

A roadmap suitable for a beginner may be unnecessarily repetitive for an experienced learner.

The challenge is therefore to use learner context before deciding the starting point.

8.3 Maintaining Logical Learning Order

Learning topics frequently have dependencies.

Advanced topics may require foundational knowledge.

A useful learning journey therefore needs to preserve a meaningful progression from foundational concepts toward more advanced capabilities.

8.4 Resource Availability

A generated learning stage does not guarantee that a suitable external resource will always be available.

PAXIS therefore treats resource discovery as an additional component rather than making resource availability a requirement for generating the learning journey.

8.5 Multilingual Interaction

Users may communicate in English, Hindi or Hinglish.

Understanding the intended goal despite differences in language and expression introduces an additional natural-language processing challenge.

8.6 Maintaining Learning State

A learner’s journey should not disappear after a single session.

The system therefore needs to preserve:

- Conversations.
- Selected journeys.
- Stage completion.
- Progress.
- Learning activity.

8.7 Determining the Next Action

Generating the initial roadmap is only one part of the problem.

The learner also needs guidance after completing a stage.

The challenge is to connect the learner's current progress with the remaining journey and determine a useful next action.

8.8 Handling Incomplete Information

Learners may not provide complete or precise information about their knowledge or goals.

The system therefore needs to operate even when the learner's initial description is not perfectly detailed.

9. Testing and Validation

9.1 Purpose of Testing

Testing PAXIS involves verifying whether the major components of the learning journey behave consistently with the learner's input.

The following areas provide meaningful validation scenarios.

9.2 Profile-Based Personalization

The system should produce different learning strategies when learners with different knowledge levels provide the same goal.

Expected Result:

The journey should reflect the learner's different starting conditions.

9.3 Goal Understanding

The system should accept naturally written learning goals.

Example:

"I want to become a Python Developer."

Expected Result:

The system should interpret the intended learning objective and generate a relevant journey.

9.4 Multilingual Input

The system should support goals expressed in English, Hindi and Hinglish.

Expected Result:

The intended learning goal should remain understandable despite the language variation.

9.5 Stage Tracking

The learner should be able to update a stage from "Not Started" to "In Progress" and subsequently to "Completed."

Expected Result:

The overall learning state should reflect the updated stage status.

9.6 Progress Tracking

The system should update progress after learning activities are completed.

Expected Result:

The learner should be able to identify their current position and completed milestones.

9.7 Next-Action Recommendation

After completion of a stage, the system should provide guidance for what should be done next.

Expected Result:

The recommended action should correspond to the learner's current journey position.

9.8 Journey Persistence

A tracked journey should remain available when the learner returns.

Expected Result:

The learner should continue from the saved state rather than beginning again.

10. Key Features Summary

The major features of PAXIS AI are summarized below.

Feature	Purpose
Personalized Journeys	Creates learning paths based on learner context and goals.
Natural-Language Interaction	Allows learners to communicate their objectives naturally.
English, Hindi and Hinglish	Supports multilingual goal communication.
Learner Profiling	Captures interests, knowledge, previous learning and available time.
Stage-Based Learning	Breaks broad objectives into manageable learning stages.
Learning Resources	Provides useful websites, videos and learning materials.
Roadmap Tracking	Allows learners to save and follow a generated journey.
Progress Tracking	Shows the learner's current completion state.
Skill Tracking	Represents development across different skill areas.
Milestones	Identifies important achievements in the learning journey.
Learning Streaks	Represents continued learning activity.
Next-Action Recommendation	Guides the learner toward the next useful activity.
Saved Conversations	Preserves previous learner interactions.
Saved Journeys	Allows previously generated roadmaps to be continued.
Downloadable Journey	Allows the learner to retain the generated roadmap.

Table 10.1: PAXIS AI Feature Summary

11. Limitations

11.1 Learner Input Dependency

The quality of personalization depends partly on the information provided by the learner.

Incomplete learner profiles can limit the amount of context available for personalization.

11.2 Ambiguous Goals

Some learning goals may be too broad or ambiguous.

For example, a statement such as “I want to learn AI” can represent many possible destinations.

Additional context may therefore be required for a more specific journey.

11.3 Resource Availability

Appropriate learning resources may not always be available for every stage.

PAXIS is designed so that the journey can still exist even when a suitable resource cannot be identified.

11.4 Progress Accuracy

Progress tracking can depend on the learner correctly updating stage completion and learning activity.

11.5 Learning Outcome

A personalized roadmap can provide structure and guidance, but the actual learning outcome also depends on learner effort, consistency, practice and completion of the recommended activities.

11.6 Model-Specific Evaluation

The available project description does not provide specific numerical evaluation results for a particular AI or machine-learning model.

Therefore, model accuracy, precision, recall or other ML performance metrics should only be reported after the corresponding implementation and evaluation have been documented.

12. Future Scope

12.1 Adaptive Journey Modification

Future versions can automatically modify the journey when the learner's performance, available time or target goal changes.

12.2 Knowledge Assessment

Initial quizzes or diagnostic assessments could be used to estimate the learner's actual knowledge instead of relying entirely on self-reported information.

12.3 Performance-Based Personalization

Quiz results, practice performance and completed activities could be used to determine whether a learner should:

- Move to the next stage.
- Revise an earlier concept.
- Perform additional practice.
- Receive a modified learning sequence.

12.4 Improved Resource Ranking

Future versions could rank resources using factors such as:

- Learner level.
- Topic relevance.
- Learning preference.
- Previous resource usage.
- Resource quality.

12.5 Interactive Practice

Learning stages could be extended with:

- Quizzes.
- Coding exercises.
- Practical assignments.
- Mini-projects.

- Assessments.

12.6 Expanded Language Support

The multilingual interaction can be extended to additional Indian and international languages.

12.7 Advanced Learning Analytics

Future versions can analyze learning patterns to identify:

- Frequently completed topics.
- Topics where learners spend more time.
- Learning consistency.
- Drop-off points.
- Skill development trends.

12.8 Long-Term Skill Profile

A long-term learner profile could maintain skill development across multiple independent learning journeys.

This would allow PAXIS to become a broader personal learning strategy system rather than a tool limited to a single goal.

13. Conclusion

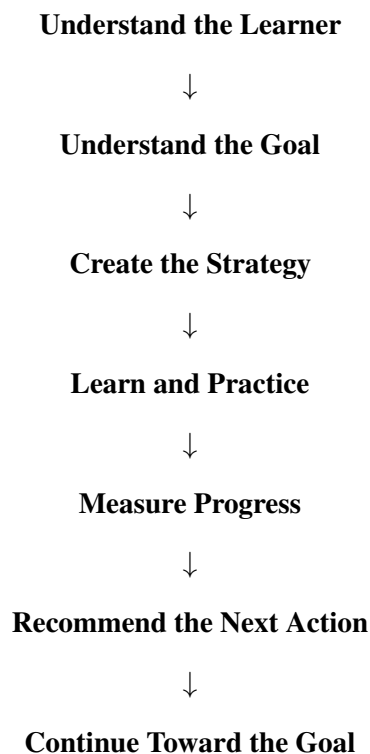
PAXIS AI presents a learner-centric approach to personalized learning journey generation.

The system begins with the learner rather than with a predefined course. It considers the learner's interests, existing knowledge, previous learning, desired outcome and available time before creating a learning strategy.

The project addresses several common problems associated with modern online learning, including uncertainty about where to begin, difficulty selecting resources, lack of structured sequencing, difficulty tracking progress and uncertainty about what to do next.

A major strength of PAXIS is the connection between the initial learning goal and the learner's continuing progress. The system does not stop after generating a roadmap. Saved journeys, stage completion, skill tracking, milestones, streaks and next-action recommendations allow the learner to continue through the journey.

The complete concept can be summarized as:



PAXIS therefore aims to change the learning experience from “searching for what to study” into “following an intelligent, personalized strategy.”

PAXIS AI

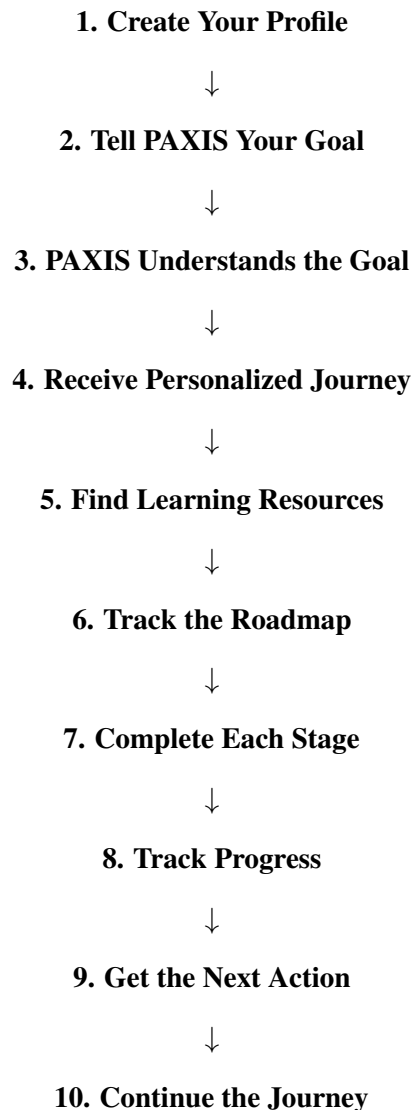
Personalized AI Exploration and Intelligent Strategy

14. References

1. PAXIS AI project description and feature specification.
2. Project README and implementation documentation, where applicable.
3. Documentation of the technologies and frameworks used in the implementation of PAXIS AI.
4. Educational resources referenced by the PAXIS learning journey system.

A. PAXIS AI Journey at a Glance

A.1 Learner Journey



A.2 PAXIS in Simple Words

You tell PAXIS where you want to go.

PAXIS understands where you are.

PAXIS creates a strategy for getting there.

You learn and practice.

PAXIS tracks your progress.

PAXIS tells you what to do next.

You continue until you reach your goal.